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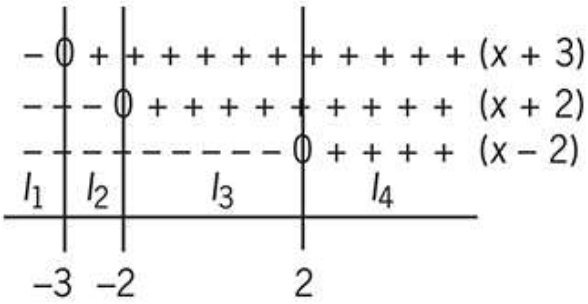


I_1: All three binomials are negative, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is NOT satisfied.

I_2: Contains no integers, so need not be considered further.

I_3: Two binomials are positive, one is negative, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is NOT satisfied.

I_4: All three binomials are positive, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is satisfied. The integers in this interval that are less than 5 are 3 and 4. (ii)

From (i) and (ii), there are 4 integers, namely –3, –2, 3, and 4, for which $\frac{(x+2)(x+3)}{x-2} \geq 0$.

The correct answer is D.

•

Of the 150 houses in a certain development, 60 percent have air-conditioning, 50 percent have a sunporch, and 30 percent have a swimming pool. If 5 of the houses have all three of these amenities and 5 have none of them, how many of the houses have exactly two of these amenities?

A. 10

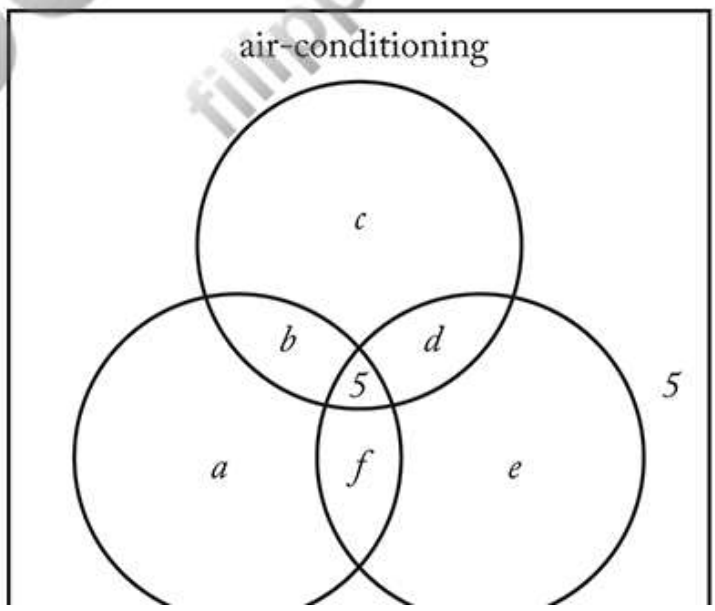
B. 45

C. 50

D. 55

E. 65

Arithmetic Sets



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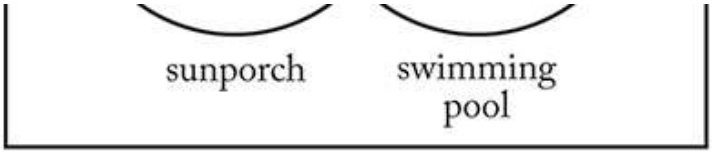

Since 60% of the 150 houses have air-conditioning, $b+c+d+5=0.6(150)=90$, so $b+c+d=85$ (i). Similarly, since 50% have a sunporch, $a+b+f+5=0.5(150)=75$, so $a+b+f=70$ (ii). Likewise, since 30% have a swimming pool, $d+e+f+5=0.3(150)=45$, so $d+e+f=40$ (iii). Adding equations (i), (ii), and (iii) gives $(b+c+d)+(a+b+f)+(d+e+f)=195$, or $a+2b+c+2d+e+2f=195$ (iv). But $a+b+c+d+e+f+5+5=150$, or $a+b+c+d+e+f=140$ (v). Subtracting equation (v) from equation (iv) gives $b+d+f=55$, so 55 houses have exactly two of the amenities.

The correct answer is D.

•

The value of $\dfrac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5}$ is how many times the value of 2^{-17}?

A. $\dfrac{3}{2}$

B. $\dfrac{5}{2}$

C. 3

D. 4

E. 5

Arithmetic Negative Exponents

If the value of $\dfrac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5}$ is x times the value of 2^{-17}, then

$$\begin{aligned}
x\left(2^{-17}\right) &= \frac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5} \\
x &= \frac{\dfrac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5}}{2^{-17}} \\
&= \frac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5} \times 2^{17} \\
&= \frac{\left(2^{-14}+2^{-15}+2^{-16}+2^{-17}\right)}{\times} \\
&= \frac{2^{-14+17}+2^{-15+17}+2^{-16+17}+2^{-17+17}}{5} \\
&= \frac{2^3+2^2+2^1+2^0}{5} \\
&= \frac{8+4+2+1}{5} \\
&= 3
\end{aligned}$$

The correct answer is C.

•

Which of the following fractions has a decimal equivalent that is a terminating decimal?

A. $\dfrac{10}{100}$

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A. $\frac{\quad}{189}$

B. $\frac{15}{169}$

C. $\frac{16}{225}$

D. $\frac{25}{144}$

E. $\frac{39}{128}$

Arithmetic Fractions; Decimals

Since the denominator of $\frac{39}{128}$, which is 128, has only factors of 2, and $\frac{1}{2} = 0.5$, the following shows that $\frac{39}{128} = 39 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$ has a terminating decimal.

$$\begin{aligned}
\frac{39}{128} &= 39 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \\
&= 19.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \\
&= 9.75 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \\
&= 4.875 \times 0.5 \times 0.5 \times 0.5 \times 0.5 \\
&= 2.4375 \times 0.5 \times 0.5 \times 0.5 \\
&= 1.21875 \times 0.5 \times 0.5 \\
&= 0.609375 \times 0.5 \\
&= 0.3046875
\end{aligned}$$

Moreover, the fact that each of the other answer choices, when expanded in this way, involves at least one factor of $\frac{1}{3} = 0.\bar{3} = 0.333\ldots$ and/or $\frac{1}{7} = 0.\overline{142857}$, neither of which has a terminating decimal expansion, suggests that none of the other answer choices has a terminating decimal.

denominator	prime factors
189	3 and 7
196	2 and 7
225	3 and 5
144	2 and 3
128	only 2

In fact, it can be proved that a fraction (numerator and denominator both integers, denominator an integer greater than 1) has a terminating decimal if and only if its denominator has only prime factors of 2 and/or 5, although this result is not needed to solve the present problem.

The correct answer is E.

•

If $\left(\frac{1}{5}\right)^m \left(\frac{1}{4}\right)^{18} = \frac{1}{2(10)^{35}}$, then $m =$

A. 17

B. 18

C. 34

D. 35

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D. 35

E. 36

Arithmetic Exponents

$\left(\frac{1}{5}\right)^m\left(\frac{1}{4}\right)^{18} = \frac{1}{2(10)^{35}}$	given
$\left(\frac{1^m}{5^m}\right)\left(\frac{1^{18}}{4^{18}}\right)^{18} = \frac{1}{2(10)^{35}}$	use $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
$\frac{1^m \times 1^{18}}{5^m \times 4^{18}} = \frac{1}{2 \times 10^{35}}$	use $\left(\frac{a}{b}\right)\left(\frac{c}{d}\right) = \frac{ac}{bd}$
$\frac{1}{5^m \times 4^{18}} = \frac{1}{2 \times 10^{35}}$	property of 1
$5^m \times 4^{18} = 2 \times 10^{35}$	take reciprocal of both sides
$5^m \times (2^2)^{18} = 2 \times 10^{35}$	$4 = 2^2$
$5^m \times 2^{36} = 2 \times 10^{35}$	use $(a^x)^y = a^{xy}$
$5^m \times 2^{36} = 2 \times (2 \times 5)^{35}$	$10 = 2 \times 5$
$5^m \times 2^{36} = 2 \times 2^{35} \times 5^{35}$	use $(ab)^x = a^x b^x$
$5^m \times 2^{36} = 2^{36} \times 5^{35}$	use $a^x a^y = a^{x+y}$
$5^m = 5^{35}$	divide both sides by 2^{36}
$m = 35$	exponents must be equal

The correct answer is D.

•

From a group of 8 volunteers, including Andrew and Karen, 4 people are to be selected at random to organize a charity event. What is the probability that Andrew will be among the 4 volunteers selected and Karen will not?

A. $\frac{3}{7}$

B. $\frac{5}{12}$

C. $\frac{27}{70}$

D. $\frac{2}{7}$

E. $\frac{9}{35}$

Arithmetic Probability; Elementary Combinatorics

The probability that Andrew will be among the 4 volunteers selected and Karen will not is equal to $\frac{A}{T}$, where A is the number of possible selections of the 4 volunteers that include Andrew and do not include Karen, and T is the total number of possible selections of the 4 volunteers. Since T is the number of combinations of 8 objects taken 4 at a time, it follows that $T = \binom{8}{4} = \frac{8!}{4!\,(8-4)!} = \frac{(5)\,(6)\,(7)\,(8)}{(1)\,(2)\,(3)\,(4)} = 70$. Since A is the number of combinations of 6 objects taken 3 at a time (select 3 volunteers from the 6 volunteers other than Andrew and Karen; including Andrew

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as a 4th selected volunteer will not change the number), it follows that

$$A = \binom{6}{3} = \frac{6!}{3!\,(6-3)!} = \frac{(4)\,(5)\,(6)}{(1)\,(2)\,(3)} = 20.$$

Therefore, the desired probability is $\frac{A}{T} = \frac{20}{70} = \frac{2}{7}$.

The correct answer is D.

•

An investor purchased 100 shares of Stock X at $6\frac{1}{18}$ dollars per share and sold them all a year later at 24 dollars per share. If the investor paid a 2 percent brokerage fee on both the total purchase price and the total selling price, which of the following is closest to the investor's percent gain on this investment?

A. 92%

B. 240%

C. 280%

D. 300%

E. 380%

Arithmetic Applied Problems; Estimation

In dollars, the investor's total cost was $100\left(6+\frac{1}{18}\right) + (2\%)100\left(6+\frac{1}{18}\right)$, which is equal to $102\left(6+\frac{1}{18}\right) = 612 + 5\frac{2}{3} \approx 618$, and the investor's total revenue was 100 (24) – (2%)100(24), which is equal to 2,400 – 48 = 2,352. Therefore, the percent gain on the investment is very nearly

$$\left(\frac{2,352-618}{618}\right) \times 100\% = \left(\frac{867}{309}\right) \times 100\%, \text{ or } \left(2+\frac{249}{309}\right) \times 100\%.$$

Since $\frac{249}{309} \approx \frac{250}{300} = \frac{5}{6} \approx 0.8$, it follows that the percent gain on the investment is closest to (2 + 0.8) × 100% = 280%.

The correct answer is C.

•

If x is the product of the integers from 1 to 150, inclusive, and 5^y is a factor of x, what is the greatest possible value of y?

A. 30

B. 34

C. 36

D. 37

E. 39

Arithmetic Properties of Integers

The greatest possible value of y such that 5^y is a factor of x is equal to the number of factors of 5 in the prime factorization of x. Because 5 is a prime number, only multiples of 5 will contribute factors of 5. Each multiple of 5 that is not a multiple of 25 contributes exactly one factor of 5, each multiple of 25 = 5 × 5 that is not a multiple of 125 contributes one additional factor of 5 beyond that already counted for multiples of 5, and 125 = 5 × 5 × 5 contributes one additional factor of 5 beyond that already counted for multiples of 25. From 1 to 150, inclusive, there are 30 multiples of 5, 6 multiples of 25, and the number 125. Therefore, the number of factors of 5 in the prime factorization of x is 30 + 6 + 1 = 37, and hence the value of y is 37.

The correct answer is D.

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The correct answer is D.

•

The number of defects in the first five cars to come through a new production line are 9, 7, 10, 4, and 6, respectively. If the sixth car through the production line has either 3, 7, or 12 defects, for which of these values does the mean number of defects per car for the first six cars equal the median?

I. 3

II. 7

III. 12

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

Arithmetic Statistics

The table shows, for each of the possible number of defects in the sixth car (shown in bold), the list of the number of defects in the first six cars in ascending order, the mean of this list, and the median of this list. For example, if the sixth car has 7 defects, then the list of the number of defects in ascending order is 4, 6, 7, 7, 9, 10; the mean of this list is the sum of these numbers divided by 6, or $\frac{43}{6}$, which is approximately 7.2; and the median of this list when in ascending order is the average of the 3rd and 4th numbers in this list, which is 7. For this example, the mean is NOT equal to the median.

list of the number of defects	mean	median
3, 4, 6, 7, 9, 10	$\frac{39}{6} = 6.5$	6.5
4, 6, 7, 7, 9, 10	$\frac{43}{6} \approx 7.2$	7
4, 6, 7, 9, 10, 12	$\frac{48}{6} = 8$	8

From the table it follows that the mean equals the median ONLY when the number of defects in the sixth car is 3 or 12.

The correct answer is D.

•

A furniture dealer purchased a desk for $150 and then set the selling price equal to the purchase price plus a markup that was 40 percent of the selling price. If the dealer sold the desk at the selling price, what was the amount of the dealer's gross profit from the purchase and the sale of the desk?

A. $40

B. $60

C. $80

D. $90

E. $100

Algebra Applied Problems; First-Degree Equations

Let S be the selling price in dollars. From the first sentence of the problem, we have $S = 150 + 0.4S$.

Thus, $0.6S = 150$ and $S = \frac{150}{0.6} = 250$. Therefore, the

selling price was $250, and hence the gross profit was $250 – $150 = $100.

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selling price was \$250, and hence the gross profit was \$250 − \$150 = \$100.

The correct answer is E.

•

How many odd numbers between 10 and 1,000 are the squares of integers?

A. 12

B. 13

C. 14

D. 15

E. 16

Arithmetic Properties of Integers

A quick way to determine which squares of integers are less than 1,000 is to observe that $30^2 = 900$ (easily done mentally) is relatively close to 1,000 and then, since odd numbers that are the squares of integers correspond exactly to the squares of odd numbers, consider the squares of odd numbers beginning with 3. This leads to the following list:

$3^2 = 9$, $5^2 = 25$,…, $31^2 = 961$, $33^2 = 1{,}089$

It follows that the odd numbers between 10 and 1,000 that are the squares of integers are 5^2, 7^2, 9^2,…, 31^2. By a direct count one finds that there are 14 squares of integers in this list. A direct count can be avoided by using the computation $\frac{31-5}{2}+1$.

The correct answer is C.

•

Three grades of milk are 1 percent, 2 percent, and 3 percent fat by volume. If x gallons of the 1 percent grade, y gallons of the 2 percent grade, and z gallons of the 3 percent grade are mixed to give $x+y+z$ gallons of a 1.5 percent grade, what is x in terms of y and z?

A. $y+3z$

B. $\frac{y+z}{4}$

C. $2y+3z$

D. $3y+z$

E. $3y+4.5z$

Algebra Applied Problems

Since the milk, when mixed, is 1.5 percent fat, the number of gallons of fat in the mixed milk is $0.015(x+y+z)$. This is also equal to the sum of the numbers of gallons of fat in the three grades of milk before they are mixed, which is $0.01x+0.02y+0.03z$, so

$$0.015(x+y+z)=0.01x+0.02y+0.03z.$$

Multiplying both sides by 1,000 (to remove the decimal fractions) and expanding the left side gives

$$15x+15y+15z=10x+20y+30z.$$

To solve for x, isolate the terms involving x on one side, combine like terms, and divide by the coefficient of x.

$$15x-10x=20y-15y+30z-15z$$

$$5x=5y+15z$$

$$x=y+3z$$

The correct answer is A.

•

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There are 8 books on a shelf, of which 2 are paperbacks and 6 are hardbacks. How many possible selections of 4 books from this shelf include at least one paperback?

A. 40

B. 45

C. 50

D. 55

E. 60

Arithmetic Elementary Combinatorics

The number of possible selections of 4 books that have at least 1 paperback is the number of possible selections of 4 books having 1 paperback and 3 hardbacks added to the number of possible selections of 4 books having 2 paperbacks and 2 hardbacks. In the case of 1 paperback and 3 hardbacks, this will be the number of combinations of 2 objects taken 1 at a time (i.e., the number of ways to choose the paperback) multiplied by the number of combinations of 6 objects taken 3 at a time (i.e., the number of ways to choose the hardbacks), which is equal to

$$\binom{2}{1} \times \binom{6}{3} = \frac{2!}{1!\,(2-1)!} \times \frac{6!}{3!\,(6-3)!} = 2 \times \frac{(4)\,(5)\,(6)}{(1)\,(2)\,(3)} = 40.$$

Similarly, the analogous calculation for the number of possible selections of having 2 paperbacks and 2 hardbacks is

$$\binom{2}{2} \times \binom{6}{2} = \frac{2!}{2!\,(2-2)!} \times \frac{6!}{2!\,(6-2)!} = 1 \times \frac{(5)\,(6)}{(1)\,(2)} = 15.$$

Thus, the number of possible selections of 4 books that have at least 1 paperback is 40 + 15 = 55.

Alternatively, determine the number of possible selections of 4 books that do NOT include at least 1 paperback and then subtract this number from the number of possible selections of 4 books without regard to how many are paperback. The number of possible selections of 4 books that do NOT include at least 1 paperback (i.e., the number of ways to choose 4 hardback books from the 6 hardback books) is the number of combinations of 6 objects taken 4 at a time, which is equal to $\binom{6}{4} = \frac{6!}{4!\,(6-4)!} = \frac{(5)}{(6)} = 15.$

The number of possible selections of 4 books without regard to how many are paperback is the number of combinations of 8 objects taken 4 at a time, which is equal to $\binom{8}{4} = \frac{8!}{4!\,(8-4)!} = \frac{(5)\,(6)\,(7)\,(8)}{(1)\,(2)\,(3)\,(4)} = 70.$

Therefore, the number of possible selections of 4 books that include at least 1 paperback is 70 – 15 = 55.

The correct answer is D.

•

If $M = \sqrt{4} + \sqrt[3]{4} + \sqrt[4]{4}$, then the value of M is

A. less than 3

B. equal to 3

C. between 3 and 4

D. equal to 4

E. greater than 4

Arithmetic Estimation; Operations on Radical Expressions

Note that $\sqrt{4} = 2$. Also, $\sqrt[3]{4} > 1$ because if $\sqrt[3]{4} \leq 1$ then $\left(\sqrt[3]{4}\right)^3 \leq 1^3$, or $4 \leq 1$, which is not true.

Similarly, $\sqrt[4]{4} > 1$. Therefore, $\sqrt{4} + \sqrt[3]{4} + \sqrt[4]{4} > 2 + 1 + 1 = 4$.

The correct answer is E.

•

When a certain tree was first planted, it was 4 feet tall, and the height of the tree increased by a constant amount each

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When a certain tree was first planted, it was 4 feet tall, and the height of the tree increased by a constant amount each year for the next 6 years. At the end of the 6th year, the tree was $\frac{1}{5}$ taller than it was at the end of the 4th year. By how many feet did the height of the tree increase each year?

A. $\frac{3}{10}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

E. $\frac{6}{5}$

Algebra Series and Sequences; First-Degree Equations

Let H be the constant amount, in feet, that the height of the tree increased each of the 6 years after the tree was planted. Since the height of the tree when planted was 4 feet, its height at the end of the 4th year was $(4+4H)$ feet and its height at the end of the 6th year was $(4+6H)$ feet. We are given that its height at the end of the 6th year was $\frac{1}{5}$ greater than its height at the end of the 4th year.

$$4+6H=\left(4+4H\right)+\frac{1}{5}\left(4+4H\right) \quad \text{given}$$

$$4+6H=\frac{24}{5}+\frac{24}{5}H \quad \text{distributive property, collect like terms}$$

$$6H=\frac{4}{5}+\frac{24}{5}H \quad \text{subtract } 4=\frac{20}{5} \text{ from both sides}$$

$$\frac{6}{5}H=\frac{4}{5} \quad \text{subtract } \frac{24}{5}H \text{ from both sides}$$

$$H=\frac{2}{3} \quad \text{divide both sides by } \frac{6}{5}$$

Alternatively, fractions can be mostly avoided by recognizing that $\frac{1}{5}$ greater is equivalent to 20% greater.

$$4+6H=1.2(4+4H) \quad \text{given}$$

$$4+6H=4.8+4.8H \quad \text{distributive property}$$

$$6H=0.8+4.8H \quad \text{subtract 4 from both sides}$$

$$1.2H=0.8 \quad \text{subtract } 4.8H \text{ from both sides}$$

$$H=\frac{0.8}{1.2}=\frac{2}{3} \quad \text{divide both sides by 1.2}$$

Therefore, the height of the tree increased by $\frac{2}{3}$ feet each year.

The correct answer is D.

•

A certain musical scale has 13 notes, each having a different frequency, measured in cycles per second. In the scale, the notes are ordered by increasing frequency, and the highest frequency is twice the lowest. For each of the 12 lower frequencies, the ratio of a frequency to the next higher frequency is a fixed constant. If the lowest frequency is 440

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frequencies, the ratio of a frequency to the next higher frequency is a fixed constant. If the lowest frequency is 440 cycles per second, then the frequency of the seventh note in the scale is how many cycles per second?

A. $440\sqrt{2}$

B. $440\sqrt{2^7}$

C. $440\sqrt{2^{12}}$

D. $440\sqrt[12]{2^7}$

E. $440\sqrt[7]{2^{12}}$

Algebra Series and Sequences

Let x be the constant ratio of the frequency of each of the 12 notes above the lowest to that of the note right below it, so that each frequency is x times the next lower one. Then the frequency $n$musical notes above the lowest note is x^n times the lowest frequency. The highest frequency is 12 musical notes above the lowest, and is also twice the lowest. Since the lowest frequency is 440 cycles per second, the highest is $440x^{12} = 440(2)$ cycles per second. Thus, $x^{12} = 2$. Therefore, since the seventh note in the scale is 6 musical notes above the lowest, the frequency of the seventh note must be $440x^6 = 440\sqrt{x^{12}} = 440\sqrt{2}$ cycles per second.

The correct answer is A.

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